Focused Architectural Brief: Multi-Agent Housing Trends & Metamorphic Testing

Course: Topics in Software Engineering (TSE)
Team Members:
- **Vijay Arvaynthan S R** (Roll: 2023111007, `vijay.s@research.iiit.ac.in`)
- **Amjad Shaik** (Roll: 2025900012, `amjad.shaik@students.iiit.ac.in`)

Target Metro: Dallas-Fort Worth, TX (DFW Core Submarkets)
Evaluation Window: 12-Month Rolling Period (2023-01 to 2023-12)
Architecture: 5 Functional Agents + 1 Formal Loop Supervisor Sentry
Verification Suite: 8 Formal Metamorphic Relations (Invariance, Monotonic, Directional)

1. Multi-Source Data Architecture & Frozen Snapshot

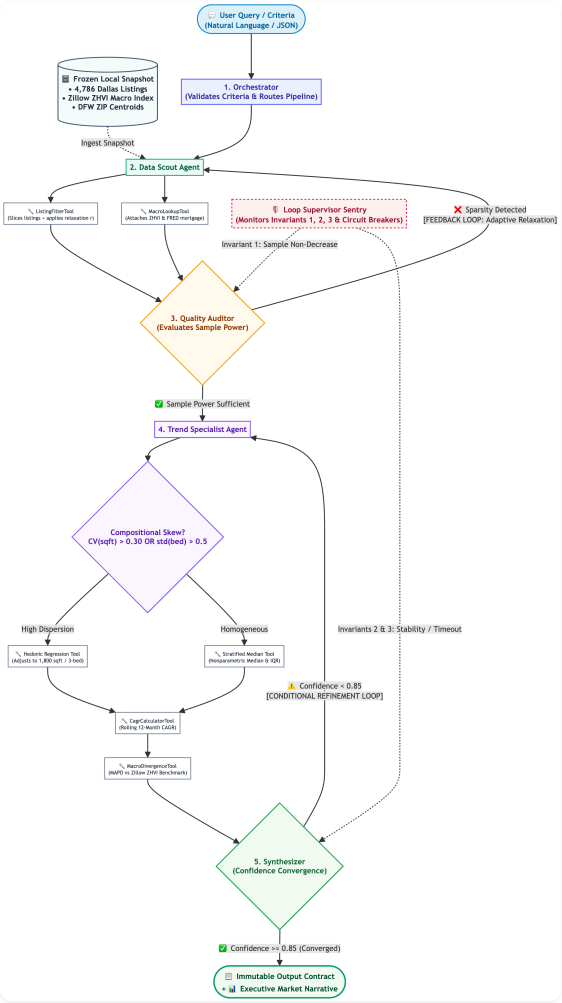
All agents operate against an immutable, frozen local data snapshot in `data/` ensuring deterministic reproducibility across test runs:

Data Asset	Path	Records / Scope	Primary Schema & Attributes	Role in Pipeline
Micro Listings	`data/dallas_listings_frozen.csv`	4,786 listings (2022-12 to 2024-01)	`listing_id`, `price`, `bedrooms`, `bathrooms`, `sqft`, `price_per_sqft`, `zipcode`, `month`, `lat`, `lon`	Ground-truth transactions harvested by Data Scout
Macro Benchmark	`data/dallas_zhvi_macro.csv`	14 monthly observations	`month`, `zhvi_median_value`, `zhvi_mom_pct`, `fred_mortgage_rate_30y`	Macroeconomic grounding & divergence scoring
Spatial Layer	`data/dallas_zip_geo.json`	10 DFW Core ZIP codes	`zipcode`, `neighborhood`, `latitude`, `longitude`, `tier`	Spatial joins & boundary relaxation mapping

Geographic Coverage. The snapshot covers 10 core Dallas residential ZIPs: Urban Condos (`75201`, `75202`), Mid-Market Residential (`75204`, `75206`, `75214`, `75219`), and Luxury Single-Family (`75205` Highland Park, `75209`, `75225` Preston Hollow, `75230`).

2. Multi-Agent Architecture & Control Flow

The architecture coordinates five specialized functional agents and one formal safety sentry:



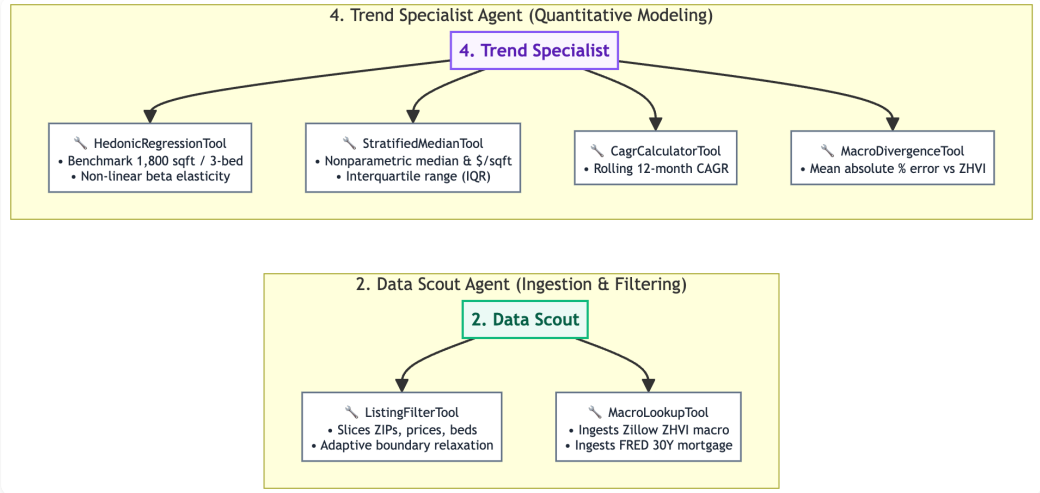
3. Agent Roster, Tool Specialization & Architectural Evaluation

3.1 Agent Responsibilities & Assignment Justification

Agent Name	Core Functional Responsibility	Assignment Mapping & Justification
1. Orchestrator	Normalizes query inputs into typed `SearchCriteria`, executes validation, manages session pipeline state.	**Pipeline entrypoint & coordination.** Guarantees valid boundary contracts before data processing.
2. Data Scout	Slices the frozen micro-listing snapshot and attaches macroeconomic benchmarks via callable tools.	**Data isolation & ingestion.** Target of upstream parameter relaxation.
3. Quality Auditor	Enforces statistical sample power ($N \geq 10$ active listings per month) across all 12 timeline bins.	**Requirement (a): Feedback loop.** Sends work back upstream to Data Scout when sample sizes fail.
4. Trend Specialist	Computes price indices, medians, dispersion spreads, and CAGR growth rates.	**Requirement (c): Genuine dependency.** Trend modeling path dynamically switches between hedonic and median methods.
5. Synthesizer	Evaluates statistical confidence and synthesizes the executive market report.	**Requirement (b): Conditional iteration.** Triggers refinement cycles if confidence < 0.85.
Loop Supervisor	External safety sentry tracking iteration counts, monotonicity, and wall-clock execution time.	**Loop safety invariants.** Enforces execution bounds (≤ 3 cycles, < 12.0 s) to prevent runaway recursion.

3.2 Tool Architecture (`backend/agents/tools.py`)

Agents delegate domain-specific calculations and data queries to 6 modular tools:



- **`ListingFilterTool`** : Executes criteria queries and applies dynamic relaxation bounds ($\pm 15\%$ price, $\pm 15\%$ sqft) during upstream feedback cycles.
- **`MacroLookupTool`** : Attaches Zillow ZHVI median levels and FRED 30-year mortgage rates for contextual alignment.
- **`HedonicRegressionTool`** : Standardizes prices against an 1,800 sqft / 3-bedroom baseline: $$P_{\text{adj}} = P_i \times \left(\frac{1800}{\text{sqft}*i}\right)^{0.75}$$ *Trigger condition: Invoked when multi-attribute skew is elevated ($CV*{\text{sqft}} > 0.30 *or* \sigma_{\text{bed}} > 0.50$); otherwise routes to* `StratifiedMedianTool` .
- **`StratifiedMedianTool`** : Nonparametric central tendency (median price, median $/sqft, IQR).
- **`CagrCalculatorTool`** : Annualized compound growth rate:

$$\text{CAGR} = \left(\frac{P_{\text{end}}}{P_{\text{start}}} \right)^{\frac{12}{n-1}} - 1$$

- **`MacroDivergenceTool`** : Mean absolute percentage deviation (MAPD) between monthly medians and Zillow ZHVI benchmark.

3.3 Extreme Sparsity Protocol ($N \leq 2$)

When rigid filters yield very few listings per month: 1. **Auditor interception:** Quality Auditor flags sparsity ($N < 10$) and triggers Data Scout with relaxation factor $r \in 0.15, 0.30, 0.45$. 2. **Supervisor termination:** Bounded at ≤ 3 cycles (< 12.0 s); halts runaway recursion if listings do not exist. 3. **Analytical fallback:** $N < 5$ diverts to `StratifiedMedianTool` . 4. **Degraded delivery:** Synthesizer assigns confidence < 0.60, sets `quality_audit_passed: false` , attaches sparsity caveats, and still delivers a valid output without crashing.

4. Formal Metamorphic Relations Catalog (8 MRs)

The metamorphic testing suite spans 8 formal relations across three canonical metamorphic families matching the assignment specification and live harness.

Output Contract Definition

Per Page 1 of the assignment specification, all MR properties evaluate against the typed 12-month output contract (`backend/core/schemas.py`):

Output Contract Field	Schema Type	Description & Mathematical Meaning
`monthly_series[m].active_count`	`int`	Number of active listings in month m (N_m)
`monthly_series[m].median_price`	`float`	Monthly median list price (quality-standardized or stratified, $)
`monthly_series[m].median_price_per_sqft`	`float`	Nonparametric median price per square foot ($/\text{sqft}$)
`annual_cagr_estimate`	`float`	12-month rolling Compound Annual Growth Rate (%)
`macro_divergence_score`	`float`	Mean absolute percentage deviation vs Zillow ZHVI (%)

4.1 Family 1: Invariance Relations

MR1: Order Invariance

- **Hypothesis:** Randomly permuting the order of raw input listings does not alter monthly medians, IQR, or CAGR.
- **Transformation:** $\mathcal{T}(D) = \text{permute}(D)$.
- **Tolerance:** $\epsilon = 10^{-5}$ (floating-point precision).
- **Target fault:** Order-dependent aggregations, unstable sorting, or index-sensitive grouping in pandas.

MR2: Duplicate Invariance

- **Hypothesis:** Duplicating every listing ($2\times$ the dataset) doubles active listing count per month, but leaves medians and $/sqft unchanged.
- **Transformation:** $\mathcal{T}(D) = D \cup D$.
- **Tolerance:** counts exact doubling; price metrics $\epsilon = 10^{-5}$.
- **Target fault:** Unweighted averaging, count inflation in median estimators, failure to handle duplicated rows correctly.

MR3: Currency Scale Invariance

- **Hypothesis:** Scaling prices from USD to USD-in-thousands ($\times 0.001$) scales medians by $0.001\times$ while MoM % changes and rankings stay the same.
- **Transformation:** $P \to 0.001 \cdot P$ (and matching criteria bounds).
- **Tolerance:** about $\$0.05$ on scaled prices; MoM $\epsilon < 10^{-3}$.
- **Target fault:** Hardcoded dollar thresholds, absolute truncation, non-homogeneous metric definitions.

4.2 Family 2: Monotonic Relations

MR4: Additive Price Monotonicity

- **Hypothesis:** Adding $+\$15,000$ to all listings in month m increases that month's median price and does not decrease it; other months stay unchanged.
- **Transformation:** $P(m) \to P(m) + 15000$.
- **Tolerance:** $\Delta\text{Median}_m \geq 0$; non-target months $\epsilon = 10^{-5}$.
- **Target fault:** Temporal leakage across monthly bins, inverted sorting, sign errors in aggregations.

MR5: Criteria Widening Monotonicity

- **Hypothesis:** Widening price criteria bounds ($P_{\min} - 60000, P_{\max} + 60000$) produces a relational superset: active listing count in follow-up is $\geq$ source count for every month.
- **Transformation:** $\mathcal{T}(\text{Criteria}) = [P_{\min} - 60000, P_{\max} + 60000]$.
- **Tolerance:** $\text{Count}{|text{follow}}(m) - |text{Count}{\text{source}}(m) \ge 0$.
- **Target fault:** Filter inversion bugs, boolean negation errors, lossy relational set intersections.

MR6: Geographic Restriction Monotonicity

- **Hypothesis:** Restricting the ZIP code search set to a proper subset produces a relational subset: active listing count in follow-up is $\leq$ source count for every month.
- **Transformation:** $\mathcal{T}(\text{ZIPs}) = \text{ZIPs}[:\max(1, |\text{ZIPs}|/2)]$.
- **Tolerance:** $\text{Count}{|text{follow}}(m) |le |text{Count}{\text{source}}(m)$.
- **Target fault:** Spatial join boundary leakage, accidental outer join fallbacks.

4.3 Family 3: Directional Relations

MR7: Window Slide Directional Invariance

- **Hypothesis:** Sliding the 12-month evaluation window forward by 1 month (`2023-01..2023-12` $\rightarrow$ `2023-02..2024-01`) preserves identical monthly statistics for the 11 overlapping months (`2023-02` through `2023-12`).
- **Transformation:** $\text{start_month} \rightarrow$ "2023-02", $\text{end_month} \rightarrow$ "2024-01".
- **Tolerance:** $\epsilon = 10^{-5}$ on overlapping monthly medians and counts.
- **Target fault:** Off-by-one temporal boundaries, timestamp parsing drift in rolling window buffers.

MR8: Multiplicative Price Scaling

- **Hypothesis:** Scaling all listing prices and criteria bounds by factor $k = 1.25$ scales monthly medians by exactly $1.25\times$, while MoM percentage changes remain invariant.
- **Transformation:** $P \rightarrow 1.25 \times P$, $P_{\min} \rightarrow 1.25 \times P_{\min}$, $P_{\max} \rightarrow 1.25 \times P_{\max}$.
- **Tolerance:** Scaled median tolerance $\pm\$0.10$; MoM % changes $\epsilon < 10^{-3}$.
- **Target fault:** Non-linear quantile estimator distortions, improper normalization in hedonic regression calculations.

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5. Statistical Metamorphic Testing Plan & Verification Protocol

Because multi-agent systems and real-world data pipelines exhibit non-linear state interactions, all relations are evaluated statistically:

1. **Base catalog:** 32 distinct search criteria slices varying across price tiers, bedroom thresholds, and ZIP codes.
2. **Fixed seeds:** Each base slice is executed under $k = 3$ fixed, recorded random seeds: $\text{SEEDS} = [42, 101, 2026]$.
3. **Statistical sample size:** $32 \text{ base slices} \times 3 \text{ seeds} \times 8 \text{ MRs} = \mathbf{768 \text{ test executions}}$.
4. **95% Wilson Score confidence interval:** Computed for the binomial violation proportion ($z = 1.96$):

$$w = \frac{p + \frac{z^2}{2n} \pm z\sqrt{\frac{p(1-p)}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

5. **One-command automated grading:** `bash export PATH="/opt/anaconda3/bin:$PATH" && python run_headline_results.py` Executes the entire 768-run suite in ≈ 7 seconds and writes results to `headline_results.json` .